



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE

Course Code & Title: EC3301 Electron Devices and Circuits

Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit I / Applications of diode
- **Course Outcome:** CO 1
- **Unit Outcome:** 1a
- **Activity Chosen:** Mind map

- **Justification:**

Mind mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mind mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches.

Time Allotted for the Activity: 05 minutes

- **Details of the Implementation:**

Mind mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mind map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

CO – PO / PSO mapping:

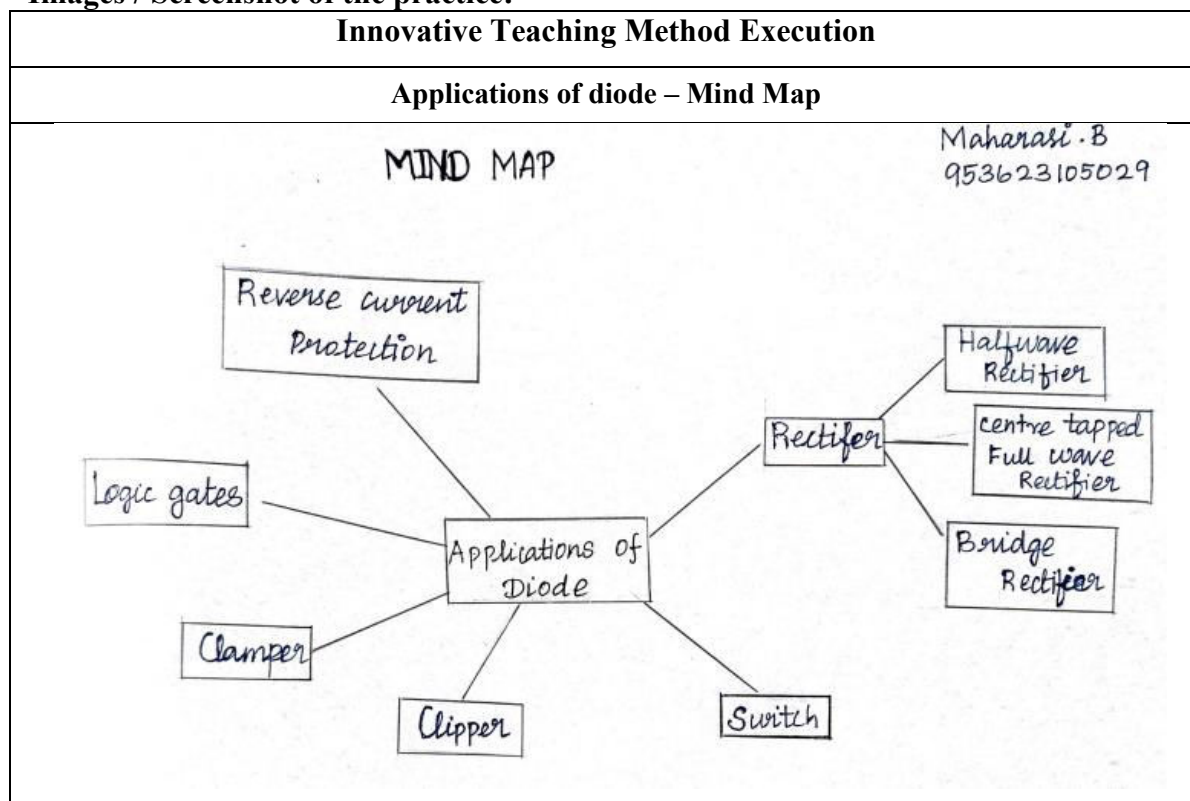
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	-	-	1	1	1	-	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- Images / Screenshot of the practice:



- Reflective Critique:

- ❖ *Feedback of practice from students and other stakeholders:*

Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ *Benefit of the practice:*

The benefits of Mind Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mind map made slow learners to remember the information more quickly.

Challenges faced in implementation:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to indicate relationships between the topics in typical load torque characteristics. I planned the activity for 05 minutes only. But in real scenario it takes 10 minutes to complete this activity.

References:

1. David A. Bell , "Electronic devices and circuits", Oxford University higher education, 5th edition 2008.
2. Sedra and smith, "Microelectronic circuits", 7th Edition., Oxford University Press, 2017

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering

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Course Code & Title: EC3301 Electron Devices and Circuits

Name of the Faculty member: Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit II / Characteristics of BJT under CE configuration
- **Course Outcome:** CO 2
- **Unit Outcome:** 2a
- **Activity Chosen:** Virtual Lab
- **Justification:**

Virtual lab refers to a virtual teaching and learning environment aimed at developing students' laboratory skills. As one of the most important eLearning tools, they allow the student to conduct various experiments without any constraints to place or time, in contrast to the constraints of real labs. By this student can understand the characteristics of BJT under CE configuration.

Time Allotted for the Activity: 15 minutes

• Details of the Implementation:

The students were asked to do the following steps

1. Initially set rheostat $R_{h1} = 1 \Omega$ and rheostat $R_{h2} = 1 \Omega$
2. Set the Collector-Emitter Voltage(V_{CE}) to 1 V by adjusting the rheostat R_{h2}
3. Base Emitter Voltage(V_{BE}) is varied by adjusting the rheostat R_{h1} .
4. Note the reading of Base current(I_B) in μ Ampere.
5. Click on 'Plot' to plot the I-V characteristics of Common-Emitter configuration. A graph is drawn with V_{BE} along X-axis and I_B along Y-axis.
6. Click on 'Clear' button to take another sets of readings
7. Now set the Collector-Emitter Voltage(V_{CE}) to 2 V, 3 V, 4 V

CO – PO / PSO mapping:

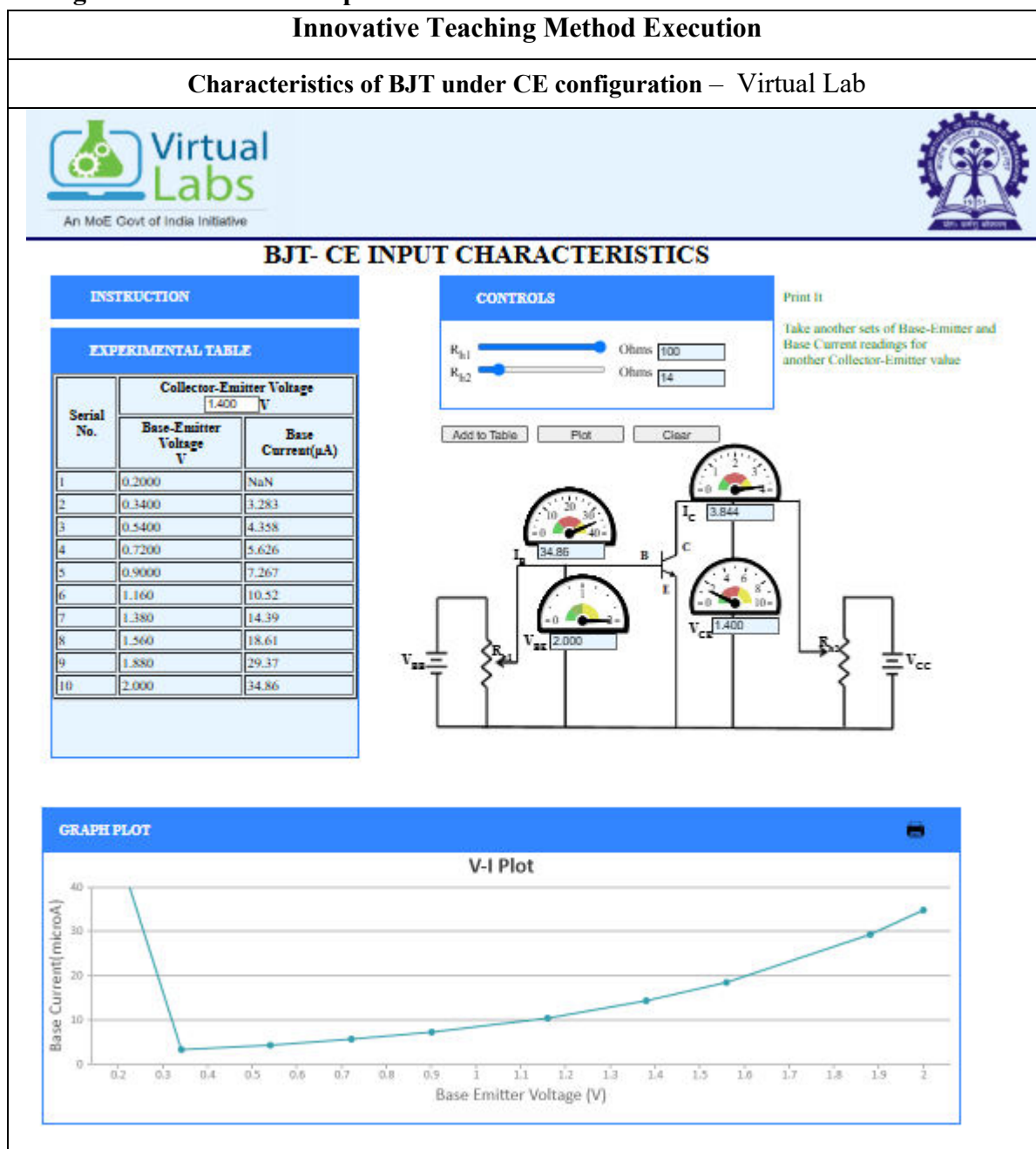
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	1	1	-	-	1	1	1	-	1	-	2	-

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO5
	1
Justification for correlation	The students can effectively use the virtual lab.

- Images / Screenshot of the practice:



- **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- ✓ Also they can able to visualise the electrostatic field, therefore they easily understand the concept.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The process of simulating a scenario to practice different responses and actions to a real life situation is extremely effective in knowledge retention. This is

because knowledge isn't in theory – the user needs to apply it in a real-life situation. Simulations can be slowed down to study behavior more closely. Conditions can be varied and outcomes investigated. Also Critical situations can be investigated without risk.

❖ *Challenges faced in implementation:*

I planned the activity for 15 minutes. But in Class room it takes 25 minutes.

References:

1. David A. Bell , "Electronic devices and circuits", Oxford University higher education, 5th edition 2008.
2. Sedra and smith, "Microelectronic circuits", 7th Edition., Oxford University Press, 2017

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Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE

Course Code & Title: EC3301 Electron Devices and Circuits

Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit III / Small signal modeling of BJT amplifiers
- **Course Outcome:** CO 3
- **Unit Outcome:** 3a
- **Activity Chosen:** One minute paper
- **Justification:**

One minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

- **Time Allotted for the Activity:** 5 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	2	2	1	1	-	-	1	1	1	-	1	-	2	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

• **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Small signal modeling of BJT amplifiers - One minute paper
S. Suvetha 953683105055 <u>ONE MINUTE PAPER</u> <u>BJT - Small Signal Model</u> The Small signal Model is a simplified representation of a transistor used to analyze its behavior for small variations around a fixed operating point (bias point). It linearizes the non-linear characteristics of the BJT, making it easier to study amplification and impedance. <u>Types of Small Signal Model :</u> 1. CE: high gain and moderate input/output impedance. 2. CB: high-frequency applications with low input impedance. 3. CC: unity gain and acts as a voltage buffer. <u>Importance of Small Signal Model :</u> Simplifies the analysis of transistor circuits. Helps determine key parameters like gain, input/output impedance, and bandwidth. Essential for designing amplifiers and small signal processing circuits.

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ Students can able to attend the question even in the questions are in indirect form.
- ✓ Students can able to explain the concepts in examination without any confusion.

❖ ***Challenges faced in implementation:***

- ✓ Time utilization for conducting activity.

References:

1. David A. Bell , "Electronic devices and circuits", Oxford University higher education, 5th edition 2008.
2. Sedra and smith, "Microelectronic circuits", 7th Edition., Oxford University Press, 2017

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Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit IV / Class C power amplifiers
- **Course Outcome:** CO 4
- **Unit Outcome:** 4e
- **Activity Chosen:** Sage and Scribe
- **Justification:**
 - ✓ The chosen topic will have the concepts related to design of various amplifiers. Hence if the students are doing co-operative learning, then it is easier for them to understand the concepts. Sage and Scribe is one such cooperative learning activity.
 - ✓ The expected outcome is like a review process of learned concepts / points, Sage and Scribe is suitable activity.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**
 - ✓ The two students sitting adjacent are made to do the activity together. One student (Sage) will explain the concept for a while and other student (Scribe) will answer for the questions.
 - ✓ For the next question, the students will change their role. Likewise it will be repeated for all the questions.
 - ✓ The students will share the same piece of paper for doing this activity.
 - ✓ I clearly explained them that, since it is a cooperative activity both is responsible for the outcome.
 - ✓ At the end of the lecture hour, the students are made to explain their answers to the class. I have given the correct answers for the questions with explanation.

CO – PO / PSO mapping:

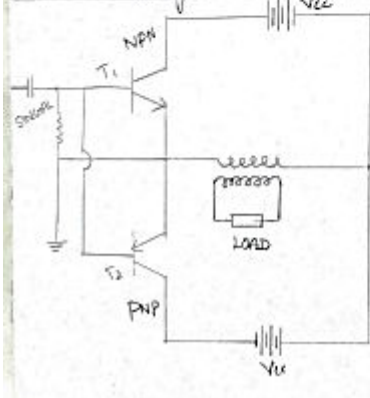
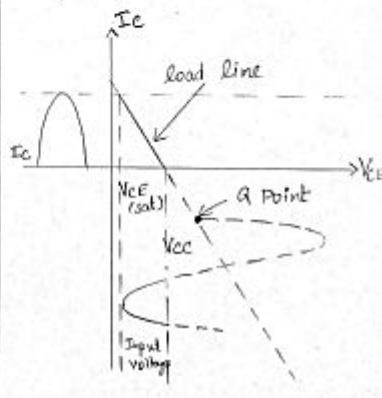
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	2	2	1	-	-	1	1	1	-	1	-	2	-

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

• Images / Screenshot of the practice:

Innovative Teaching Method Execution	
Class C power amplifiers – Sage and Scribe	
SAGE AND SCRIBE CLASS C POWER AMPLIFIER	
D. Karishma	M. Mahalakshmi
<u>Introduction</u> ¹ • If the collector current flows for less than half-cycle of the input signal, it is called class C power amplifier. • In Class C amplifier, the base is given some negative bias so that collector current does not flow just when the positive half-cycle of the signal starts. Such amplifiers are never used for power amplification. • However, they are used as tuned amplifiers i.e. to amplify a narrow-band of frequency near the resonant frequency.	<u>Working</u> ³ • The class C amplifier is a nonlinear amplifier because the output is not a replica of the input. • Although it is not used in linear applications, it finds applications in high-frequency sine wave generators and RF oscillators. For this reason, the load is a resonant circuit in which a common-emitter amplifier is used. During the short interval, the transistor is turned on.
<u>Diagram</u> ² 	<u>Waveform</u> ⁴ 

Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- ✓ The students have liked the session and they felt comfortable in doing this activity, since they are interacting with their neighbor. They did the work easily.
- ✓ They conveyed that more such activity will help them in understanding the concepts clearly and easily.

❖ **Benefit of the practice:**

- ✓ The students have understood the concepts clearly and it has been evident from the answers they have written on the answer paper.
- ✓ The students have presented the answers to the class. It cleared any doubt in understanding the important points in the chosen topic.

❖ **Challenges faced in implementation:**

- ✓ In few groups, both the students were slow learners; hence they find it very difficult to answer the questions. I helped them in answering the questions by providing them with simple hints.
- ✓ In the next time, while doing this activity, I should properly pair the students.

References:

1. David A. Bell , "Electronic devices and circuits", Oxford University higher education, 5th edition 2008.
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Innovative Practice Description

- **Unit / Topic:** Unit V / Oscillators
- **Course Outcome:** CO 5
- **Unit Outcome:** 5b
- **Activity Chosen:** Class Poll
- **Justification:**

Class Poll is an evidence-based teaching practice that can be a great way to actively engage students in their own learning. Instructors can use polls to have students make predictions about course content, stimulate class discussion with peers, and generate new ideas. Instructors at Cornell are using classroom polling in many creative ways to help students become active critical thinkers.

- **Time Allotted for the Activity:** 05 minutes
- **Details of the Implementation:**

The poll questions that required students to do more than remember previously encountered information. Also the poll questions were posted at particular interval of teaching, so student can active throughout the class. The poll is displayed on the projector and asked them to raise their hand and discussed about the answer.

CO – PO / PSO mapping:

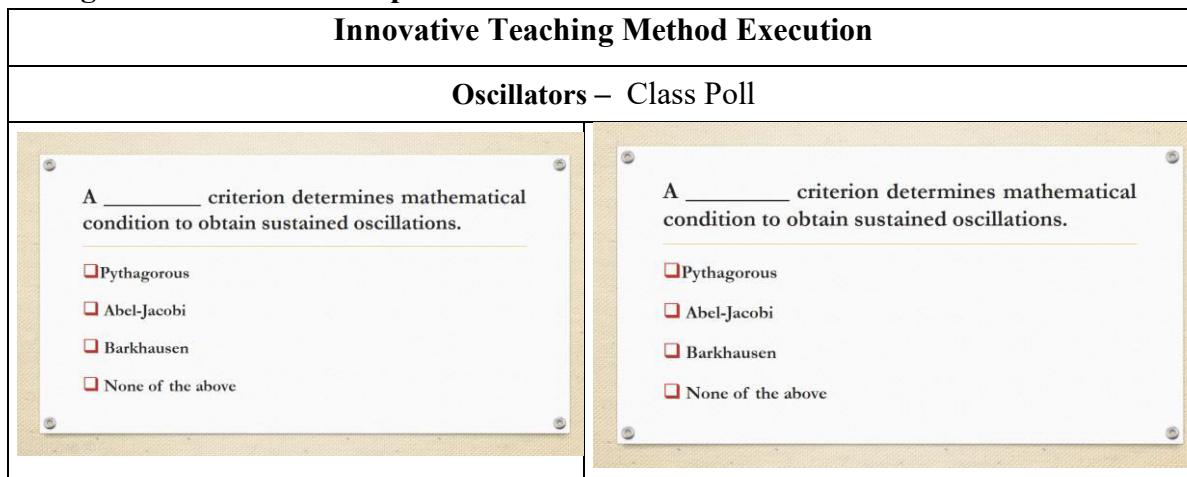
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	2	2	1	-	-	1	1	1	-	1	-	2	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

• **Images / Screenshot of the practice:**



Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- ✓ The students have liked the session and they felt comfortable in doing this activity. They did the work easily.
- ✓ They conveyed that more such activity will help them in understanding the concepts clearly and easily.

❖ **Benefit of the practice:**

- ✓ The students have understood the concepts clearly and it has been evident from the answers they have written on the answer paper.
- ✓ The students have presented the answers to the class. It cleared any doubt in understanding the important points in the chosen topic.

❖ **Challenges faced in implementation:**

- ✓ the slow learners were felt very difficult to answer the questions. I helped them in answering the questions by providing them with simple hints.
- ✓ In the next time, while doing this activity, I should properly pair the students.

References:

1. David A. Bell , "Electronic devices and circuits", Oxford University higher education, 5th edition 2008.
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